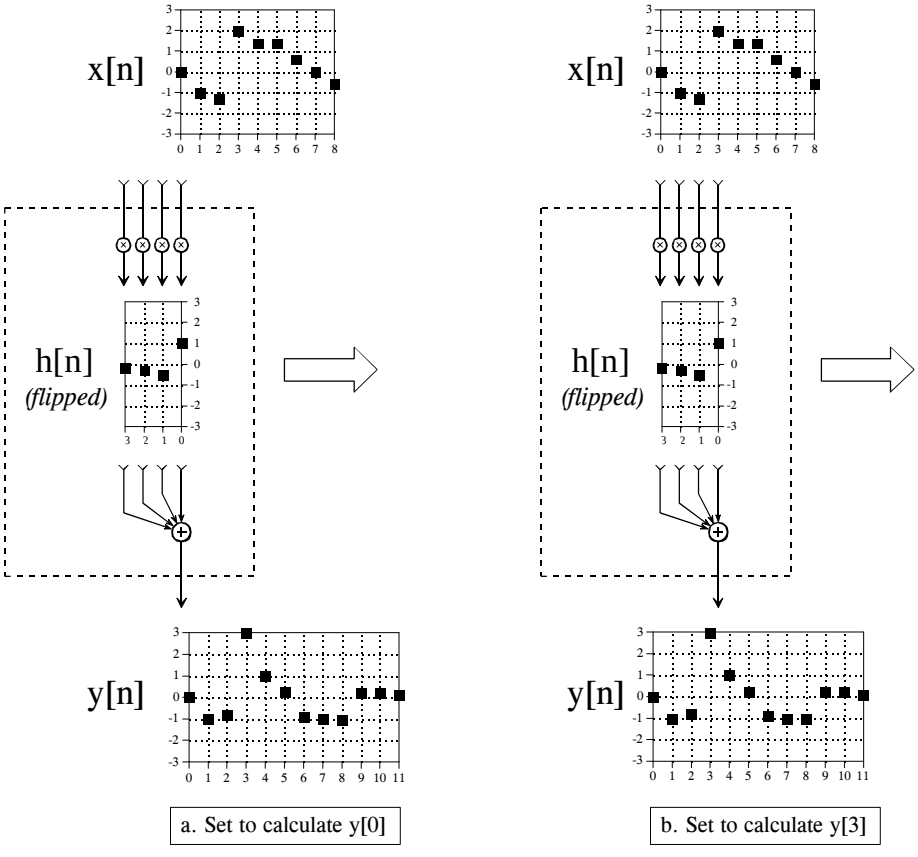

FIGURE 6-9
The convolution machine in action. Figures (a) through (d) show the convolution machine set to calculate four different output signal samples, y[0], y[3], y[8], and y[11].

Figure 6-9 shows the convolution machine being used to calculate several samples in the output signal. This diagram also illustrates a real nuisance in convolution. In (a), the convolution machine is located fully to the left with its output aimed at $y[0]$. In this position, it is trying to receive input from samples: $x[-3]$, $x[-2]$, $x[-1]$, and $x[0]$. The problem is, three of these samples: $x[-3]$, $x[-2]$, and $x[-1]$, do not exist! This same dilemma arises in (d), where the convolution machine tries to accept samples to the right of the defined input signal, points $x[9]$, $x[10]$, and $x[11]$.

One way to handle this problem is by *inventing* the nonexistent samples. This involves adding samples to the ends of the input signal, with each of the added samples having a value of *zero*. This is called **padding** the signal with zeros. Instead of trying to access a nonexistent value, the convolution machine receives a sample that has a value of zero. Since this zero is eliminated during the multiplication, the result is mathematically the same as *ignoring* the nonexistent inputs.